STREGONI / B TO BERN

Retrieval works.
The evidence is inspectable.

HelioSense proof of concept

A standalone fallback for demonstrating backend retrieval without a working frontend connection.

Test: 19 October 2026, 09:22 UTC / 11:22 Madrid

Capture searched	Demo queries	Passages returned	Distinct documents
127,902 chunks	4 completed	120 occurrences	52 documents

What the test demonstrates

The production hybrid-retrieval adapter returned real institutional passages for the fictional HelioSense demo. Selected passages identify relevant people and explain sensing, closed-loop and glucose-prediction capabilities. Every returned passage matched its exact stored body slice and body hash.

Named source-backed examples

Lilian Witthauer - sensing/diabetes technology; passage rank 3 in the lay-wording query.
José García-Tirado - closed-loop systems; passage rank 9 in that query.
Stavroula Mougiakakou - historical algorithms research; passage rank 19 in the sensor query.

Scope of this proof

This demonstrates document retrieval and inspected researcher mentions. The accepted structured-entity projection returned zero profiles for every query. It is not a complete expert-directory or Recall@10 benchmark. Historical publications require freshness review.

Fictional company input / real corpus evidence / no frontend / no database writes

01 / The demo and measured results

HelioSense is a fictional Swiss medtech company exploring wearable continuous glucose monitoring. It asks for Bern expertise in sensing, algorithms, device development and evaluation. No real company plan was submitted.

Input	Search time	Hits	Literal checks
full-demo-brief	24.824 s	30	30 / 30 passed
glucose-sensors	21.839 s	30	30 / 30 passed
cgm	18.119 s	30	30 / 30 passed
biosensing-lay-terms	17.542 s	30	30 / 30 passed

Times cover encoding/search before separate body verification and entity projection. Each input was run once. Cold/warm conditions were not controlled.

full-demo-brief

The entire demo/case-brief.md was submitted verbatim, including simulation boundaries. The exact input is preserved in the companion JSON.

glucose-sensors

Which UniBE, ARTORG or Diabetes Center Berne groups or researchers have documented expertise relevant to glucose sensors or continuous glucose monitoring for an exploratory wearable medtech R&D project? Show exact sources and distinguish direct from adjacent matches.

cgm

For a fictional wearable medtech R&D scenario, find source-backed Bern expertise relevant to CGM (continuous glucose monitoring). Include people, units, projects and services only where the UniBE, ARTORG or Diabetes Center Berne pages support them; explain missing fields.

biosensing-lay-terms

A fictional startup wants a wearable that measures blood sugar continuously. Who or what in the UniBE, ARTORG and Diabetes Center Berne ecosystem has documented work on biosensing, wearable measurements, diabetes technology or related device evaluation? Link every factual match to its source and label adjacent work clearly.

Procedure

Queries contained no researcher names and used no name-based prefilter. Lexical and dense branches ran independently, then reciprocal rank fusion combined them. No reranker or web discovery was used. Examples were selected only after inspecting results.

All 30 passages per query are retained in the companion rank/source record. A passage's rank is not the rank of a deduplicated person. No expert-level score is inferred.

02 / Source-backed people

Lilian Witthauer

Direct sensing research lead in a 2023 source

The retrieved interview identifies Witthauer with diabetes technology at UniBE/DCB and describes a continuous blood-glucose sensor. This supports an exploratory sensing lead for HelioSense.

> Lilian Witthauer ist Assistenzprofessorin für Diabetestechnologie an der Universität Bern und dem Diabetes Center Berne (DCB).
>
> Eines meiner aktuellen Projekte be - fasst sich mit der Entwicklung eines Sensors, der kontinuierlich und verzö- gerungsfrei den Glukosegehalt im Blut misst.

biosensing-lay-terms, rank 3 | Open original institutional source

Chunk fda246c6-c0da-446d-9fd8-00ed9856c9a8 | body characters 3022-5422

Boundary: Historical 2023 interview. Current role, technology readiness and availability require confirmation.

José García-Tirado

Adjacent systems/control capability in a 2024 source

The DCB publication lists García-Tirado with closed-loop systems. The same passage describes glucose sensors, insulin pumps and algorithmic control. This is a documented systems connection, not proof of a HelioSense-compatible sensor.

> Prof. Dr José García-Tirado, Systèmes en boucle fermée
>
> technologiques composés de capteurs de glucose et de pompes à insuline,

biosensing-lay-terms, rank 9 | Open original institutional source

Chunk 270952d8-c4fc-4a8a-8517-3d79c113d443 | body characters 0-2400

Boundary: 2024 source context; no claim of a verified current appointment or willingness to collaborate.

Original source language/characters are retained; whitespace is normalized for display. Exact text and absolute source character offsets are preserved in retrieval-results.json.

03 / Historical lead and capability evidence

Stavroula Mougiakakou

Historical diabetes algorithms lead; follow-up verification needed

The ARTORG report names Mougiakakou in Diabetes Technology Research and discusses glucose-profile prediction with statistical/computational methods. This supports historical relevance to signal processing.

> Diabetes Technology Research Stavroula Mougiakakou, Technical Head of Research Group
>
> Data-driven models, based on advanced statistical and computational intelligence meth- ods are developed for the prediction of glucose profile in the near future

glucose-sensors, rank 19 | Open original institutional source

Chunk 1907f038-4573-4df7-ab3f-2e456f534ce9 | body characters 78777-80753

Boundary: The URL identifies a 2012 report. This is historical evidence, not a current role/contact listing.

Direct capability from the full demo brief

> For the glucose sensors, different optical methods are combined that measure the glucose concentration directly in the blood.

full-demo-brief, rank 2 | Open original institutional source

Chunk ea4ab916-d993-446c-9731-b872cece3bc3 | body characters 1053-1716

The DCB Sensing & Smart Monitoring passage supports optical glucose-measurement research without assigning that capability to a person not named in this passage.

Ranking still needs improvement

The sensor-wording query ranked a DextEgg fine-motor rehabilitation page first. The full-brief query ranked a generic ARTORG passage first. These are weak matches, not supported glucose-sensing claims. Need-fit assessment and source inspection remain necessary.

04 / Verify or replay without the frontend

Mechanical checks

For each of 120 returned passage occurrences, the ordinary authorized connection read the stored document version, compared body[char_start:char_end] with the hit and recomputed the complete body's SHA-256. All passed. Literal integrity does not establish semantic correctness or source truth.

Connections enforced read-only transactions and bounded statement/lock timeouts. An existing owned capture was reused. No case, snapshot, claim, grant, corpus update or migration was created.

Item	Recorded value
Schema	0012_membership_lookup
Snapshot	658224cd-9e05-4fa1-bbbb-bc7d309f7fdf
Build	15ea2e50-d169-4ab0-a652-13a6df0df633
Encoder	google/embeddinggemma-2; 768 dimensions
Encoder revision	914f7f89142e33e77833254d9c9b90c3cef7303b
Application source	a35d70af7f65 + existing local changes
Strategy	Independent lexical/dense + RRF; no reranker
Evidence record	docs/proof-of-concept/retrieval-results.json

Replay using a private authorized environment and an existing owned run:

```
python -m backend.evaluation.proof_of_concept --env-file PRIVATE_ENV --run-id OWNED_RUN_ID
```

Optional --db-host HOST --db-port PORT selects the same authorized database's connection address. Credentials and local tokenizer assets are not included. Full instructions and hashes are in the companion README.

What remains unproven

Independent human gold labels, expert precision/Recall@10, current availability, complete profiles, UniBE-outside-ARTORG coverage, full Apertus reasoning/drafting, API export, frontend integration and target deployment were not tested. These are inspectable research leads, not invitations, clinical advice or partnerships.

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